Week 8

$$f(x) = x^{\sqrt{x}} \qquad x > 0$$

find $f'(x)$ let $y = x^{\sqrt{x}}$

Apply ln on both sides:

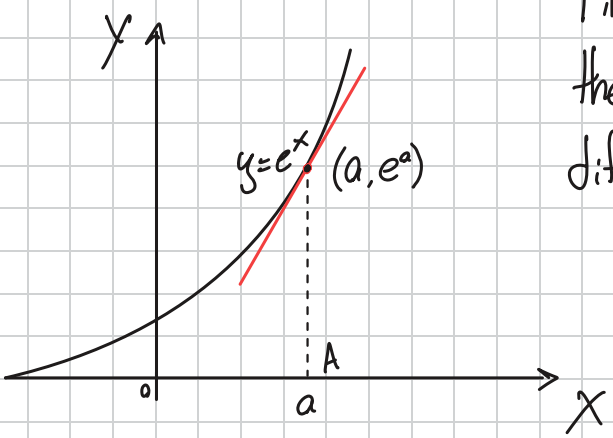
$$\ln y = \ln\left(x^{\sqrt{x}}\right) = \sqrt{x}\ln x$$

$$\ln y = \sqrt{x}\ln x$$

$$\frac{y'}{y} = \frac{1}{2\sqrt{x}} \cdot \ln x + \sqrt{x} \cdot \frac{1}{x} = \frac{\ln x + 2}{2\sqrt{x}}$$

$$y' = x^{\sqrt{x}} \frac{\ln x + 2}{2\sqrt{x}}$$

Find equation of the tangent line to $y = e^x$ that passes through the origin



Find the slope of the line in two different ways

1. Geometry: in triangle OAP

Slope $m = \frac{PA}{OA} = \frac{e^a}{a}$

2. Derivative: $m = y'(x = a)$

$y'(x) = e^x \rightarrow m = e^a$

the two numbers must be equal

$\frac{e^a}{a} = e^a$

Multiply both sides by a

$\frac{1}{a} = 1 \rightarrow \boxed{a = 1}$

$\rightarrow$ Slope $m = e$

$\rightarrow$ $y = mx + b$

Here $b = 0$ (line passes through origin)

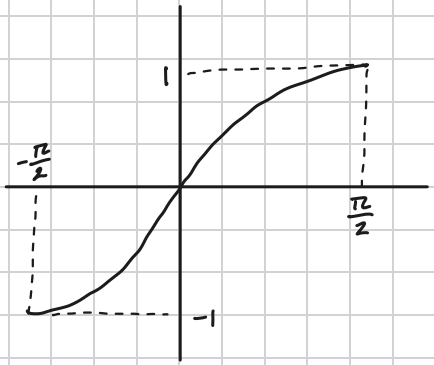
$\Rightarrow y = ex$

inverse trig

$y = \sin x$

$D = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$R = [-1, 1]$



invertable

$f^{-1}(x) = \sin^{-1}(x) = \arcsin x$

$D(\arcsin x) = [-1, 1]$

$R(\arcsin) = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}(\sin x) = x \quad , \quad x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\sin(\sin^{-1} x) = x \quad , \quad x \in [-1, 1]$

$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \quad \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$

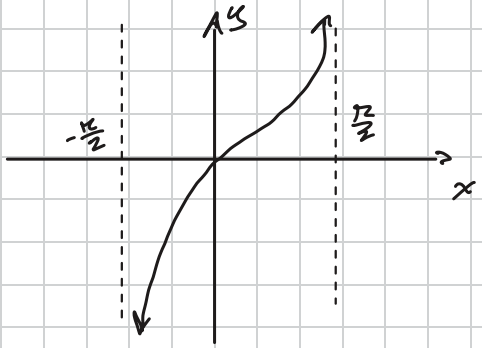
Ex $a > 0$

$$\frac{d}{dx}\left(\sin^{-1}\frac{x}{a}\right) = \frac{1}{\sqrt{1-\frac{x^2}{a^2}}}\cdot\frac{1}{a} = \frac{1}{\sqrt{\frac{a^2-x^2}{a^2}}}\cdot\frac{1}{a}$$

$$= \frac{\cancel{a}}{\sqrt{a^2-x^2}}\cdot\frac{1}{\cancel{a}} = \frac{1}{\sqrt{a^2-x^2}}$$

$$\Rightarrow \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$y = f(x) = \tan x$



$D(\tan x) = (-\frac{\pi}{2}, \frac{\pi}{2})$

$R(\tan x) = \mathbb{R}$

invertible

$f^{-1}(x) = \tan^{-1} x = \arctan x$

$D(\tan^{-1}) = \mathbb{R}$

$R(\tan^{-1}) = (-\frac{\pi}{2}, \frac{\pi}{2})$

$\tan^{-1}(\tan x) = x \quad , \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$\tan(\tan^{-1} x) = x \quad , \quad x \in \mathbb{R}$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} \Rightarrow \int \frac{dx}{1+x^2} = \tan^{-1} x + c$$

Solve the IVP: $\begin{cases} y' = \frac{1}{4+x^2} \\ y(0) = 3 \end{cases}$

$$y = \int \frac{dx}{4+x^2} = \frac{1}{2}\tan^{-1}\frac{x}{2} + c$$

$a = 2$

$$y(0) = c = 3 \longrightarrow y(x) = \frac{1}{2}\tan^{-1}\frac{x}{2} + 3$$